April Lyu

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EDUCATION

UCLA ANDERSON SCHOOL OF MANAGEMENT Los Angeles, CA
Master of Financial Engineering Dec 2024
Relevant Coursework: Derivatives, Fixed Income, Financial Risk Management, Data Science & Machine Learning, Asset Pricing

NEW YORK UNIVERSITY New York, NY
B.A., Economics and Mathematics / Computer Science May 2023

WHAT I CAN BRING

- **Technical Skills**: Java, Python, R, SQL, Excel, Tableau, and Alteryx
- Experienced in monitoring portfolio risk exposures and conducting quantitative research; skilled in automating and streamlining financial modeling, transforming large datasets into actionable insights with Python to enhance efficiency

EXPERIENCE

UCLA ANDERSON SCHOOL OF MANAGEMENT – FINANCIAL ENGINEERING PROGRAM Los Angeles, CA
Research Assistant Feb 2025 – Present

- Conducted literature reviews in financial economics and econometrics; summarized findings into model specifications and collected large financial/macro datasets using SQL, Python, and R
- Developed and validated econometric models (regression, time-series, factor) with robustness and sensitivity testing
- Produced publication-ready tables, charts, and figures for manuscripts and presentations; replicated results and maintained audit trails to ensure accuracy, reproducibility, and research integrity

BOHAI SECURITIES Tianjin, China
Quantitative Investment Intern Jun 2024 – Sep 2024

- Completed replication of four quantitative research projects, independently managing data acquisition, cleansing, factor calculation, analysis, and report generation; optimized code and models to reduce complexity and improve accessibility
- Researched market timing strategies, evaluating factor sensitivity and validity with varying thresholds and signal windows for beta-hedging to mitigate market downside risk of investment. Developed several market-driven factors and conducted statistical tests to create risk-neutral portfolio strategies, enhancing portfolio risk management through long-short positioning
- Built a dynamic model to track investment returns for one of China's largest companies, integrating APIs and statistical methods to detect trading movements, enabling periodic updates and continuous monitoring of the company's investments

Evolution Global Advisors San Diego, CA
Part-time Remote Financial Analyst Intern Jun 2024 – Aug 2024

- Selected stocks based on portfolio requirements and conducted comprehensive research on chosen stocks, and performed market trend and sector-specific analysis to identify industry trends and benchmark companies against peers
- Employed valuation models to estimate intrinsic value, built technical charts and quantitative tools for security analysis, and delivered research reports and presentations with commentary in weekly meetings to support portfolio adjustments

FIRSTRADE SECURITIES New York, NY
Project Management Department Intern March 2023 – Jul 2023

- Collected information through 50+ hours of competitor research and analysis, delivering proposals with insights derived from data analytics to leadership for developing ideas in enhancing operational strategies and optimizing resource allocation
- Utilized Python to process and analyze raw data, ensuring data was effectively structured and comprehensible to use as supplements for decision support and performance monitoring in the manager's daily meetings with the CEO

PROJECTS

Statistics Series Course - Econometrics and Empirical Analysis with R - Data Scientist Focus Jan 2024 – March 2024

- Applied a spectrum of econometric models in R, showcasing proficiency in causal inference and statistical analysis
- Performed empirical analysis using advanced models such as ARMA, GARCH, and Fama-French in R. Conducted autocorrelation tests, and trend analysis to select optimal models, and derived insights into predictability, heteroscedasticity, and underlying data dynamics

Financial Risk Management Course-Quantitative Risk Modeling Project March 2024 – Jun 2024

- Calculated VaR for major banks (2006–2008) using parametric and bootstrap methods, validating results through back-testing and sensitivity testing; applied EWMA and GARCH models to capture time-varying volatility in financial time series
- Conducted credit risk modeling and related analysis, including credit loss forecasting. Calculated default probabilities and recovery rates on debt using the Merton model to estimate credit risk

Convertible Bond Pricing and Arbitrage Analysis Project Nov 2024 – Dec 2024

- Developed valuation models for convertible bonds using a replicating portfolio approach and a binomial lattice framework. Performed data collection, cleaning, and exploration of Bloomberg market data to simulate option prices using a Monte Carlo model, addressing data limitations of the replication approach. Incorporated real-time inputs such as stock prices and volatility into the lattice model while accounting for credit spreads and dilution effects
- Spearheaded sensitivity analysis on strike price adjustments to assess their impact on pricing accuracy and replication performance, identifying mispricing and arbitrage opportunities to support risk-hedging strategies